

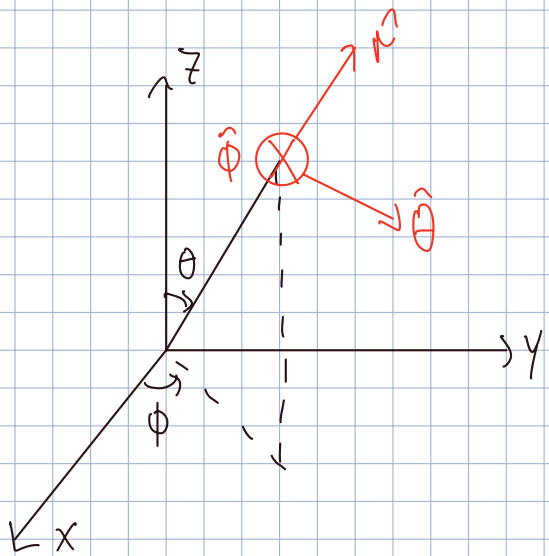
PHYS 486 - HW07

1) Recall that $\vec{L} = \vec{r} \times \vec{p}$

$\vec{p} = -i\hbar \frac{\partial}{\partial \mathbf{x}}$ simply in 1-D but in 3D we should replace $\frac{\partial}{\partial \mathbf{x}}$ by the gradient operator $\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$

$$\hat{L} = -i\hbar (\hat{r} \times \hat{\nabla})$$

$$\text{where } \hat{\nabla} = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$



$$\hat{r} \times \hat{\theta} = \hat{\phi} \quad \& \quad \hat{r} \times \hat{\phi} = -\hat{\theta}$$

$$\hat{\phi} = \hat{y} \cos \phi - \hat{x} \sin \phi$$

$$\hat{\theta} = \hat{x} \cos \phi \cos \theta + \hat{y} \sin \phi \cos \theta - \hat{z} \sin \theta$$

$$\hat{r} = \hat{z} \cos \theta + \hat{x} \cos \phi \sin \theta + \hat{y} \sin \phi \sin \theta$$

$$\vec{r} \times \vec{\nabla} = \hat{\phi} \frac{\partial}{\partial \theta} - \hat{\theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \Rightarrow \hat{L} = -i\hbar \left(\hat{\phi} \frac{\partial}{\partial \theta} - \hat{\theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right)$$

Replacing $\hat{\phi}$, $\hat{\theta}$ with their cartesian expressions, $\vec{L}$ can be written as

$$\vec{L} = \hat{L}_x \hat{x} + \hat{L}_y \hat{y} + \hat{L}_z \hat{z} \quad \text{where}$$

$$\hat{L}_x = i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y = i\hbar \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$\begin{aligned}
\hat{L}^2 &= \hat{L} \cdot \hat{L} = -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} - \hat{\phi} \frac{\partial}{\partial \theta} \left(\hat{\theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right) - \hat{\theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \left(\hat{\phi} \frac{\partial}{\partial \theta} \right) \right. \\
&\quad \left. + \hat{\theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \left(\hat{\theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right) \right] \\
&= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} - \hat{\phi} \left(\frac{\partial \hat{\theta}}{\partial \theta} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} + \hat{\theta} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right) \right) - \hat{\theta} \frac{1}{\sin \theta} \left(\frac{\partial \hat{\phi}}{\partial \phi} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{\partial}{\partial \phi} \frac{\partial}{\partial \theta} \right) \right. \\
&\quad \left. + \hat{\theta} \frac{1}{\sin \theta} \left(\frac{\partial \hat{\theta}}{\partial \phi} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} + \hat{\theta} \frac{1}{\sin \theta} \frac{\partial^2}{\partial \phi^2} \right) \right]
\end{aligned}$$

Notice that $\frac{\partial \hat{\phi}}{\partial \theta} = 0$, $\frac{\partial \hat{\phi}}{\partial \phi} = -\cos \theta \hat{\theta} - \sin \theta \hat{\phi}$, $\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$, $\frac{\partial \hat{\theta}}{\partial \phi} = \cos \theta \hat{\phi}$

$$\begin{aligned}
\hat{L}^2 &= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} - \hat{\phi} \left[-\hat{r} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} + \hat{\theta} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right) \right] - \hat{\theta} \frac{1}{\sin \theta} \left(-\cos \theta \frac{\partial}{\partial \theta} \hat{\theta} + \hat{\phi} \frac{\partial^2}{\partial \phi \partial \theta} \right) \right. \\
&\quad \left. + \frac{\hat{\theta}}{\sin \theta} \left(\cos \theta \hat{\phi} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} + \hat{\theta} \frac{1}{\sin \theta} \frac{\partial^2}{\partial \phi^2} \right) \right] \\
&= -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\
&= -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]
\end{aligned}$$

→ Now let us write the Hamiltonian in spherical coordinates

$$H = \hat{T} + V(r) = -\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \underbrace{\frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}}_{\frac{\hat{L}^2}{r^2}} \right] + V(r)$$

→ Notice that angular part of kinetic energy is directly prop. to $\hat{L}^2$ (as it should since $\hat{T} = \hat{T}_{\text{angular}} + \hat{T}_{\text{radial}}$). $\hat{L}^2$ constitutes the kinetic energy due to the rotation (in classical mechanics).

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\hat{T}_{\text{radial}}(r) + \frac{\hat{L}^2(\theta, \phi)}{r^2} \right) + V(r)$$

Now using Heisenberg's equation of motion

$$\frac{d\langle \hat{L} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{L}] \rangle \quad \Rightarrow \quad \text{Note that } \frac{\partial \hat{L}}{\partial t} \text{ term is 0 since } \hat{L} \text{ doesn't explicitly depend on time.}$$

→ Now realize that all the terms in the Hamiltonian (except $\hat{L}^2$) is a function of r only and $\hat{L}$ doesn't depend on r at all. Operators of separate orthogonal directions commute! Therefore

$$[\hat{L}(\theta, \phi), \hat{T}_{\text{radial}}(r)] = [\hat{L}(\theta, \phi), V(r)] = 0 \quad (\text{also } [L^2, L] = 0 \text{ trivially})$$

$$\text{Thus } [\hat{L}, \hat{H}] = 0 \quad \Rightarrow \quad \frac{d\langle \hat{L} \rangle}{dt} = 0$$

Alternative Solution

Let us analyze $\langle [\hat{H}, \hat{L}_i] \rangle$ where $i \in \{x, y, z\}$

$$\langle [\hat{H}, \hat{L}_i] \rangle = \int \psi^* (\hat{H} \hat{L}_i \psi) dV - \int \psi^* (\hat{L}_i \hat{H} \psi) dV$$

↳ Since we know $V(r)$ commutes with $\hat{L}_i$ algebraically, let us focus on the Laplacian term ∇^2 .

$$\langle [\hat{H}, \hat{L}_i] \rangle = \int \psi^* (\nabla^2 \hat{L}_i \psi) dV - \int \psi^* (\hat{L}_i \nabla^2 \psi) dV$$

Now Apply Green's 2nd identity with $f = \psi^*$, $g = \hat{L}_i \psi$

$$\int_U (f \nabla^2 g - g \nabla^2 f) dV = \oint_{\partial U} (f \nabla g - g \nabla f) \cdot d\vec{S}$$

$$\int \psi^* \nabla^2 (\hat{L}; \psi) dV - \int \hat{L}; \psi \nabla^2 (\psi^*) dV = \text{surface integral.}$$

We can take the surface integral through a large sphere with $r \rightarrow \infty$. Since all well-behaved wavefunction and their derivatives must vanish at infinity term inside the surface integral goes to zero for our large sphere. Therefore,

$$\int \psi^* \nabla^2 (\hat{L}; \psi) dV - \int \hat{L}; \psi \nabla^2 (\psi^*) dV = 0$$

↳ Using Hermitian property of $\hat{L}_i$, manipulate second integral to get

$$\int \psi^* \nabla^2 (\hat{L}; \psi) dV - \int \psi^* \hat{L}_i (\nabla^2 \psi) dV = 0$$

Thus, we arrive finally at $\langle [\hat{L}, \hat{H}] \rangle = 0 \Rightarrow \frac{d\langle \hat{L} \rangle}{dt} = 0$

↳ Note that $\langle [\hat{L}_i, \hat{H}] \rangle = 0$ for all x, y, z . So that yields $\langle [\hat{L}, \hat{H}] \rangle$.

$$2) \quad Y_{0,0} = \sqrt{\frac{1}{4\pi}}, \quad Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos\theta, \quad Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin\theta e^{\pm i\phi}$$

2.a) 2 wave functions are orthogonal if $\langle \psi_1 | \psi_2 \rangle = 0$

Inner prod in 3D coordinates is

$$\langle \psi_1 | \psi_2 \rangle = \int_0^\infty \int_0^\pi \int_0^{2\pi} \psi_1^* \psi_2 r^2 \sin\theta dr d\theta d\phi$$

For spherical-harmonics $\langle Y_{l',m'} | Y_{l,m} \rangle = \int_0^{2\pi} \int_0^\pi Y_{l',m'}^* Y_{l,m} \sin\theta d\theta d\phi$
 Since they're already normalized on r ,

$$\langle Y_{0,0} | Y_{1,0} \rangle = \frac{\sqrt{3}}{4\pi} \int_0^{2\pi} \left[\int_0^\pi \sin\theta \cos\theta d\theta \right] d\phi = 0$$

$$\frac{1}{2} \int_0^\pi \sin 2\theta d\theta = \frac{1}{4} \left[\cos 2\theta \right]_0^\pi = 0$$

$$\langle Y_{0,0} | Y_{1,\pm 1} \rangle = \mp \sqrt{\frac{3}{32\pi^2}} \int_0^\pi \left[\int_0^{2\pi} e^{\pm i\phi} d\phi \right] \sin^2\theta d\theta = 0$$

$$\left[\frac{e^{\pm i\phi}}{\pm i} \right]_0^{2\pi} = \frac{1-1}{\pm i} = 0$$

$$\langle Y_{1,0} | Y_{1,\pm 1} \rangle = \pm \sqrt{\frac{9}{32\pi^2}} \left[\int_0^\pi \sin\theta \cos\theta d\theta \right] \left[\int_0^{2\pi} e^{\pm i\phi} d\phi \right] = 0$$

→ Now let us check the normalization

$$\langle Y_{0,0} | Y_{0,0} \rangle = \frac{1}{4\pi} \left[\int_0^\pi \sin\theta d\theta \right] \left[\int_0^{2\pi} d\phi \right] = \frac{1}{4\pi} \underbrace{[-\cos\theta]_0^\pi}_{2} \cdot 2\pi = 1$$

$$\langle Y_{1,0} | Y_{1,0} \rangle = \frac{3}{4\pi} \left[\int_0^\pi \sin\theta \cos^2\theta d\theta \right] \left[\int_0^{2\pi} d\phi \right] = \frac{3}{2} \int_0^\pi \cos^2\theta \sin\theta d\theta$$

Let $\left. \begin{array}{l} u = \cos\theta \\ du = -\sin\theta d\theta \end{array} \right\} \langle Y_{1,0} | Y_{1,0} \rangle = -\frac{3}{2} \int_1^{-1} u^2 du = \frac{3}{2} \left[\frac{u^3}{3} \right]_{-1}^1 = 1$

$$\begin{aligned} \langle Y_{1,\pm 1} | Y_{1,\pm 1} \rangle &= \frac{3}{8\pi} \left[\int_0^\pi \sin^3\theta d\theta \right] \left[\int_0^{2\pi} \underbrace{e^{-i\phi} e^{\pm i\phi}}_1 d\phi \right] = \frac{3}{4} \int_0^\pi \sin^3\theta d\theta \\ &= \frac{3}{4} \int_0^\pi (1 - \cos^2\theta) \sin\theta d\theta = \frac{3}{4} \left[\underbrace{\int_0^\pi \sin\theta d\theta}_2 - \underbrace{\int_0^\pi \cos^2\theta \sin\theta d\theta}_{2/3 \text{ from prev. options}} \right] = \frac{3}{4} \left(2 - \frac{2}{3} \right) = 1 \end{aligned}$$

→ We see that spherical harmonics form an orthonormal basis.

2.b) First let's look at $\hat{L}^2$

- $\hat{L}^2 Y_{0,0} = 0$ since it $\sqrt{\frac{1}{4\pi}}$ is merely a constant.

↳ $\hat{L}^2 Y_{0,0} = 0 \cdot Y_{0,0} \Rightarrow Y_{0,0}$ is an eigenfunc. of $\hat{L}^2$ with eigval. 0.

- $\hat{L}^2 Y_{1,0} = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \left(\sqrt{\frac{3}{4\pi}} \cos\theta \right) \right) \right] \Rightarrow$ since no ϕ dependence of $Y_{1,0}$
 $= -\hbar^2 \sqrt{\frac{3}{4\pi}} \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} (-\sin^2\theta) \right] = \hbar^2 \sqrt{\frac{3}{4\pi}} \frac{1}{\sin\theta} (2 \sin\theta \cos\theta)$
 $= 2\hbar^2 \sqrt{\frac{3}{4\pi}} \cos\theta = 2\hbar^2 Y_{1,0}$

↳ $Y_{1,0}$ is an eigenfunc. of $\hat{L}^2$ with eigval $2\hbar^2$

$$\begin{aligned}
 \bullet \hat{L}^2 Y_{1,\pm 1} &= -\hbar^2 \sqrt{\frac{3}{8\pi}} \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} (\sin\theta \cos\theta e^{\pm i\phi}) - \frac{1}{\sin^2\theta} \sin\theta e^{\pm i\phi} \right] \Rightarrow \frac{\partial^2 e^{\pm i\phi}}{\partial\phi^2} = (\pm i)^2 e^{\pm i\phi} = -e^{\pm i\phi} \\
 &= -\hbar^2 \sqrt{\frac{3}{8\pi}} \left[\frac{\cos^2\theta - \sin^2\theta - 1}{\sin\theta} \right] e^{\pm i\phi} = -\hbar^2 \sqrt{\frac{3}{8\pi}} \left(\frac{-2\sin^2\theta}{\sin\theta} \right) e^{\pm i\phi} \\
 &= 2\hbar^2 \sqrt{\frac{3}{8\pi}} \sin\theta
 \end{aligned}$$

$\hookrightarrow Y_{1,\pm 1}$ is an efunc. of $\hat{L}^2$ with eigval $2\hbar^2$,

Now do the same thing for $\hat{L}_z = -i\hbar \frac{\partial}{\partial\phi}$

$$\bullet \hat{L}_z Y_{0,0} = 0 Y_{0,0} \Rightarrow Y_{0,0} \text{ is an eigfunc. of } \hat{L}_z \text{ with eigval } 0.$$

since constant

$$\bullet \hat{L}_z Y_{1,0} = 0 Y_{1,0} \Rightarrow Y_{1,0} \text{ is an eigfunc. of } \hat{L}_z \text{ with eigval } 0.$$

$\downarrow$
 $Y_{1,0}$ depends only on θ
 and not ϕ

$$\begin{aligned}
 \bullet L_z Y_{1,\pm 1} &= -i\hbar \frac{\partial}{\partial\phi} \left(\mp \sqrt{\frac{3}{8\pi}} \sin\theta e^{\pm i\phi} \right) = -i\hbar \sqrt{\frac{3}{8\pi}} \sin\theta (\mp) (\pm i) e^{\pm i\phi} \\
 &= (-i\hbar) (\pm i) \left(\mp \sqrt{\frac{3}{8\pi}} \sin\theta e^{\pm i\phi} \right) = (\pm \hbar) \left(\mp \sqrt{\frac{3}{8\pi}} \sin\theta e^{\pm i\phi} \right)
 \end{aligned}$$

$\hookrightarrow Y_{1,\pm 1}$ is an eigfunc of $\hat{L}_z$ with eigval $\pm\hbar$,

2.c) Recall $L_x = -i\hbar \left[-\sin\varphi \frac{\partial}{\partial\theta} - \cos\varphi \cot\theta \frac{\partial}{\partial\phi} \right]$, $L_z = -i\hbar \frac{\partial}{\partial\phi}$

$$L_y = -i\hbar \left[\cos\varphi \frac{\partial}{\partial\theta} - \sin\varphi \cot\theta \frac{\partial}{\partial\phi} \right]$$

$$L_{\pm} = L_x \pm i L_y = -i\hbar \left[(-\sin\varphi \pm i \cos\varphi) \frac{\partial}{\partial\theta} - (\cos\varphi \pm i \sin\varphi) \cot\theta \frac{\partial}{\partial\phi} \right]$$

$$L_{\pm} = \pm\hbar e^{\pm i\varphi} \left[\frac{\partial}{\partial\theta} \pm i \cot\theta \frac{\partial}{\partial\phi} \right] \Rightarrow \text{raising / lowering operators}$$

→ We know that highest possible state for a given l is $Y_{l,l}$ since $m = \{-l, \dots, l\}$. Raising it any higher would destroy the state.

$$L_+ Y_{l,l} = \hbar e^{i\phi} \left[\frac{\partial Y_{l,l}}{\partial \theta} + i \cot \theta \frac{\partial Y_{l,l}}{\partial \phi} \right] = 0$$

→ Since separable write $Y_{l,l}(\theta, \phi) = P(\theta) e^{il\phi}$. Then

$$\frac{dP(\theta)}{d\theta} e^{il\phi} + i \cot \theta (il) P(\theta) e^{il\phi} = 0$$

→ ϕ dependence of spherical harmonics is always $e^{il\phi}$

$$\frac{dP(\theta)}{P(\theta)} = l \frac{\cos \theta}{\sin \theta} d\theta = l \frac{d(\sin \theta)}{\sin \theta}$$

→ Integrating both sides $\ln P(\theta) = l \ln(\sin \theta) + C \Rightarrow P(\theta) = N \sin^l(\theta)$

$$Y_{l,l} = N_l \sin^l \theta e^{il\phi}$$

→ Let's normalize to find the constant N_l

$$\langle Y_{l,l} | Y_{l,l} \rangle = |N_l|^2 \int_0^{2\pi} \underbrace{e^{-il\phi} e^{il\phi}}_{2\pi} d\phi \int_0^\pi \sin^{2l+1} \theta d\theta$$

One can find $\int_0^\pi \sin^{2l+1} \theta d\theta = \frac{2^{2l+1} (l!)^2}{(2l+1)!}$ via Gamma/Beta functions

$$N_l = \frac{1}{2^l \cdot l!} \sqrt{\frac{(2l+1)!}{4\pi}} \Rightarrow Y_{l,l} = \frac{1}{2^l \cdot l!} \sqrt{\frac{(2l+1)!}{4\pi}} \sin^l \theta e^{il\phi}$$

$$Y_{16,16} = \frac{1}{2^{16} \cdot 16!} \sqrt{\frac{33!}{4\pi}} \sin^{16} \theta e^{i16\phi}$$

d) Similarly we will have $L_- Y_{l,-l} = 0$

Following the same step we will obtain the same differential eq. since minus sign from $\mp i \cot \theta$... and from derivative $\frac{\partial Y}{\partial \phi}$ will cancel.

We will obtain the same $P(\theta)$ but angular part is $e^{-il\phi}$

$$Y_{l,-l} = \frac{1}{2^l l!} \sqrt{\frac{(2l+1)!}{4\pi}} \sin^l \theta e^{il\phi}$$

$$Y_{16,-16} = \frac{\sqrt{33!/4\pi}}{2^{21}} \sin^{16} \theta e^{-i16\phi}$$

Now use the raising operator 1 time to arrive at $Y_{16,-15}$
Remember the identity (proved in discussion 8)

$$\hat{L}_{\pm} |l, m\rangle = \hbar \sqrt{l(l+1) - m(m\pm 1)} |l, m\pm 1\rangle$$

$$\hat{L}_{+} |16, -16\rangle = \hbar \sqrt{16 \cdot 17 - (-16)(-15)} |16, -15\rangle = \hbar \sqrt{32} |16, -15\rangle$$

but also $L_{\pm} = \pm \hbar e^{\pm i\varphi} \left[\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right]$

$$\begin{aligned} \hat{L}_{+} |16, -16\rangle &= \hbar e^{i\varphi} \left(\frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) (N_{16} \sin^{16} \theta e^{-i16\varphi}) \\ &= \hbar e^{i\varphi} N_{16} (16 \sin^{15} \theta \cos \theta e^{-i16\varphi} + 16 \cot \theta \sin^{16} \theta e^{-i16\varphi}) \\ &= 32 \hbar N_{16} \sin^{15} \theta \cos \theta e^{-i15\varphi} \end{aligned}$$

Then equating those 2 expressions for $\hat{L}_{+} |16, -16\rangle$, we get

$$Y_{16,-15} = 4\sqrt{2} N_{16} \sin^{15} \theta \cos \theta e^{-i15\varphi} = \boxed{\frac{4\sqrt{2}}{2^{16} \cdot 16!} \sqrt{\frac{33!}{4\pi}} \sin^{15} \theta \cos \theta e^{-i15\varphi}}$$

3.a) Recall $\hat{L}_{\pm} |l, m\rangle = \hbar \sqrt{l(l+1) - m(m\pm 1)} |l, m\pm 1\rangle$

$$\langle L_{\pm} \rangle = \langle l, m | L_{\pm} | l, m \rangle = \hbar \sqrt{l(l+1) - m(m\pm 1)} \langle l, m | l, m\pm 1 \rangle$$

Since we've shown that spherical harmonics are orthonormal

$$\langle l, m | l, m\pm 1 \rangle = 0$$

Then $\langle L_{\pm} \rangle = 0$. Therefore we obtain

$$\left. \begin{array}{l} L_+ = L_x + iL_y \\ L_- = L_x - iL_y \end{array} \right\} \begin{array}{l} \langle L_x \rangle = \frac{1}{2} (\langle L_+ \rangle + \langle L_- \rangle) = 0 \\ \langle L_y \rangle = \frac{1}{2i} (\langle L_+ \rangle - \langle L_- \rangle) = 0 \end{array}$$

3.b) Using identities $\hat{L}^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle$
 $\hat{L}_z |l, m\rangle = \hbar m |l, m\rangle$

Since $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \Rightarrow \langle \hat{L}_x^2 \rangle + \langle \hat{L}_y^2 \rangle = \langle \hat{L}^2 \rangle - \langle \hat{L}_z^2 \rangle$

$$\begin{aligned} \langle \hat{L}_x^2 \rangle + \langle \hat{L}_y^2 \rangle &= \hbar^2 l(l+1) \langle l, m | l, m \rangle - \hbar^2 m \langle l, m | l, m \rangle \\ &= \hbar^2 (l(l+1) - m^2) \end{aligned}$$

→ Now let us try to find a relation between $\langle \hat{L}_x^2 \rangle$, $\langle \hat{L}_y^2 \rangle$

$$L_x^2 = \left[\frac{1}{2} (L_+ + L_-) \right]^2 = \frac{1}{4} (L_+^2 + L_-^2 + L_+ L_- + L_- L_+)$$

$$L_y^2 = \left[\frac{1}{2i} (L_+ - L_-) \right]^2 = \frac{1}{4} (-L_+^2 - L_-^2 + L_+ L_- + L_- L_+)$$

Take the expectation values of L_x^2 , L_y^2 but first realize $\langle L_+ \rangle = \langle L_- \rangle = 0$

This is trivial either from the previous option or the fact that

$$\langle l, m | L_{\pm}^2 | l, m \rangle = \langle l, m | l, m \pm 2 \rangle = 0 \Rightarrow \text{orthogonal states}$$

However $\langle L_+ L_- \rangle = \langle L_- L_+ \rangle \neq 0$ since raising followed by lowering brings it to the same state (possibly with scaling)

Therefore we obtain, $\langle L_x^2 \rangle = \langle L_y^2 \rangle = \frac{1}{4} (\langle L_+ L_- \rangle + \langle L_- L_+ \rangle)$

$$\langle \hat{L}_x^2 \rangle = \langle \hat{L}_y^2 \rangle = \frac{1}{2} [\langle \hat{L}_x^2 \rangle + \langle \hat{L}_y^2 \rangle] = \boxed{\frac{\hbar^2}{2} [l(l+1) - m^2]}$$

3.C) *Let us prove $\Delta L_x \Delta L_y \geq \frac{\hbar}{2} |\langle L_z \rangle|$ first

$$(\Delta L_x)^2 = \langle L_x^2 \rangle - \underbrace{\langle L_x \rangle^2}_0 = \frac{\hbar^2}{2} [l(l+1) - m^2]$$

$$(\Delta L_y)^2 = \langle L_y^2 \rangle - \underbrace{\langle L_y \rangle^2}_0 = \frac{\hbar^2}{2} [l(l+1) - m^2]$$

$$\langle \hat{L}_z \rangle = \langle l, m | \hat{L}_z | l, m \rangle = \langle l, m | m\hbar | l, m \rangle = m\hbar$$

$$(\Delta L_x) (\Delta L_y) = \frac{\hbar^2}{2} [l(l+1) - m^2], \quad \frac{\hbar}{2} |\langle L_z \rangle| = \frac{\hbar^2}{2} |m|$$

Notice that since $l \geq |m|$, $l(l+1) \geq |m|(|m|+1) = m^2 + |m|$

$$l(l+1) - m^2 \geq |m| \Rightarrow \text{Thus } (\Delta L_x)(\Delta L_y) \geq \frac{\hbar}{2} |\langle L_z \rangle|$$

→ Now let's tackle $\Delta L_y \Delta L_z \geq \frac{\hbar}{2} |\langle L_x \rangle|$

$$\left. \begin{aligned} \langle L_z^2 \rangle &= m^2 \hbar^2 \Rightarrow \Delta L_z = m^2 \hbar^2 - (m\hbar)^2 = 0 \\ \langle L_x \rangle &= 0 \end{aligned} \right\} 0 \geq 0 \text{ satisfied } \checkmark$$

→ Similarly $\Delta L_z \Delta L_x \geq \frac{\hbar}{2} |\langle L_y \rangle|$

↳ $\Delta L_z = 0$ and $\langle L_y \rangle = 0$ satisfied ✓

* For 2 components of uncertainty to be 0, we need $\Delta L_x = 0$ or $\Delta L_y = 0$ since $\Delta L_z = 0$ already.

$$(\Delta L_{x,y})^2 = \langle L_{x,y}^2 \rangle - \langle L_{x,y} \rangle^2 = \frac{\hbar^2}{2} [l(l+1) - m^2] = 0$$

We require $l(l+1) = m^2$

↳ but $l \geq |m| \Rightarrow l^2 + l \geq m^2$

↳ Only way two sides can be equal is if $m = l = 0$

* Uncertainties vanish for $Y_{0,0}$ state. This makes absolute sense since $Y_{0,0}$ is spherically symmetric and has no angular momentum to be uncertain at all ∇_0